

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا،

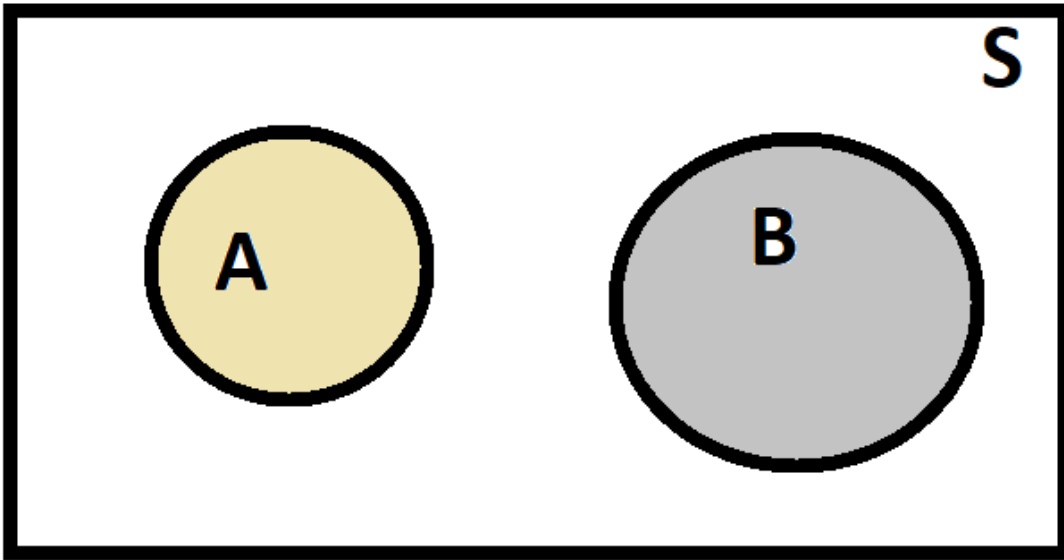
وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا،

The Additive and Multiplication Rule

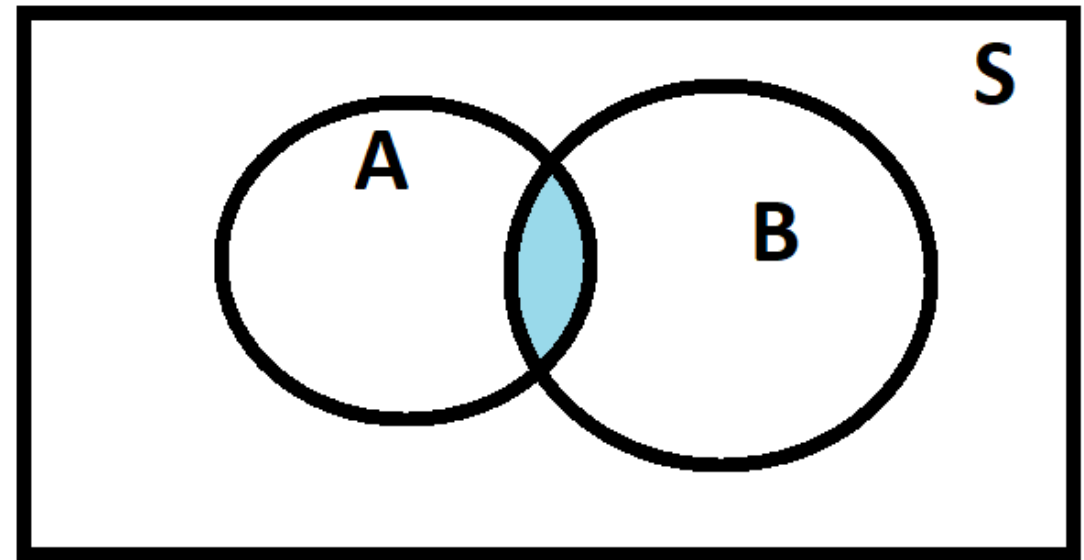
Week 4

Events

Union: $A \cup B$



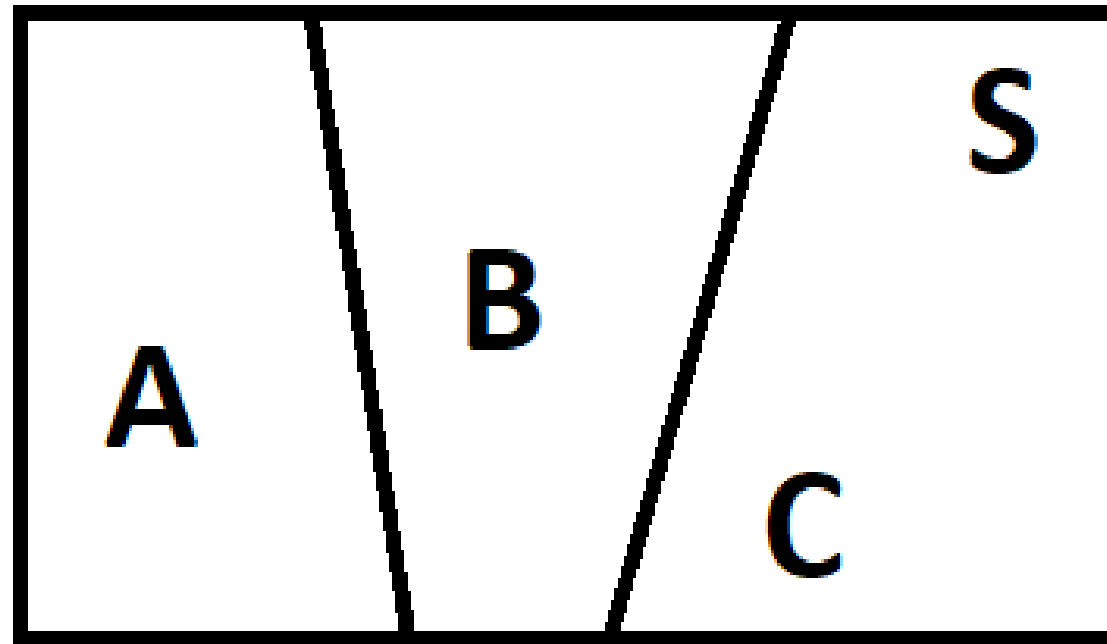
Intersection: $A \cap B$



Events

Mutually exclusive/ disjoint: $A \cap B = \varnothing$

Collectively Exhaustive: $A \cup B \cup C = S$



De Morgan's Law

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

Example: 1

In describing the state of supply of construction materials, if E_1 represent the shortage of concrete, E_2 represent shortage of steel, then

1. Shortage of construction material means:

$$E_1 \cup E_2$$

2. No shortage of construction material means:

$$(E_1 \cup E_2)'$$

3. Shortage of both materials means:

$$E_1 \cap E_2$$

Example: 2

The transportation of cargoes between Peshawar and Karachi may be by air (A), highway (H) or railway (R), then

1. Cargoes may be shipped by air or highway or railway means:

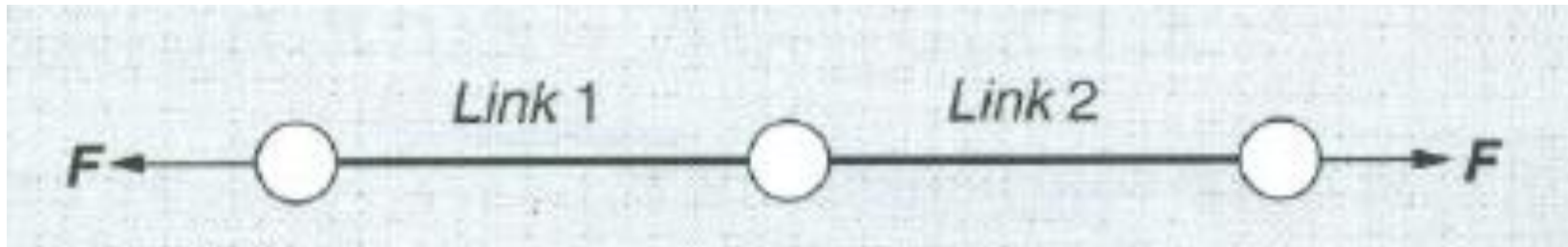
$$A \cup H \cup R$$

2. Cargoes may be shipped by air alone means:

$$A \cap H' \cap R'$$

Example: 3

Consider a simple chain of two links as in fig. Let E_1 be the breakage of link 1 and E_2 be the breakage of link 2.



1. Failure of chain means:

$$E_1 \cup E_2$$

2. No failure of chain means:

$$(E_1 \cup E_2)'$$

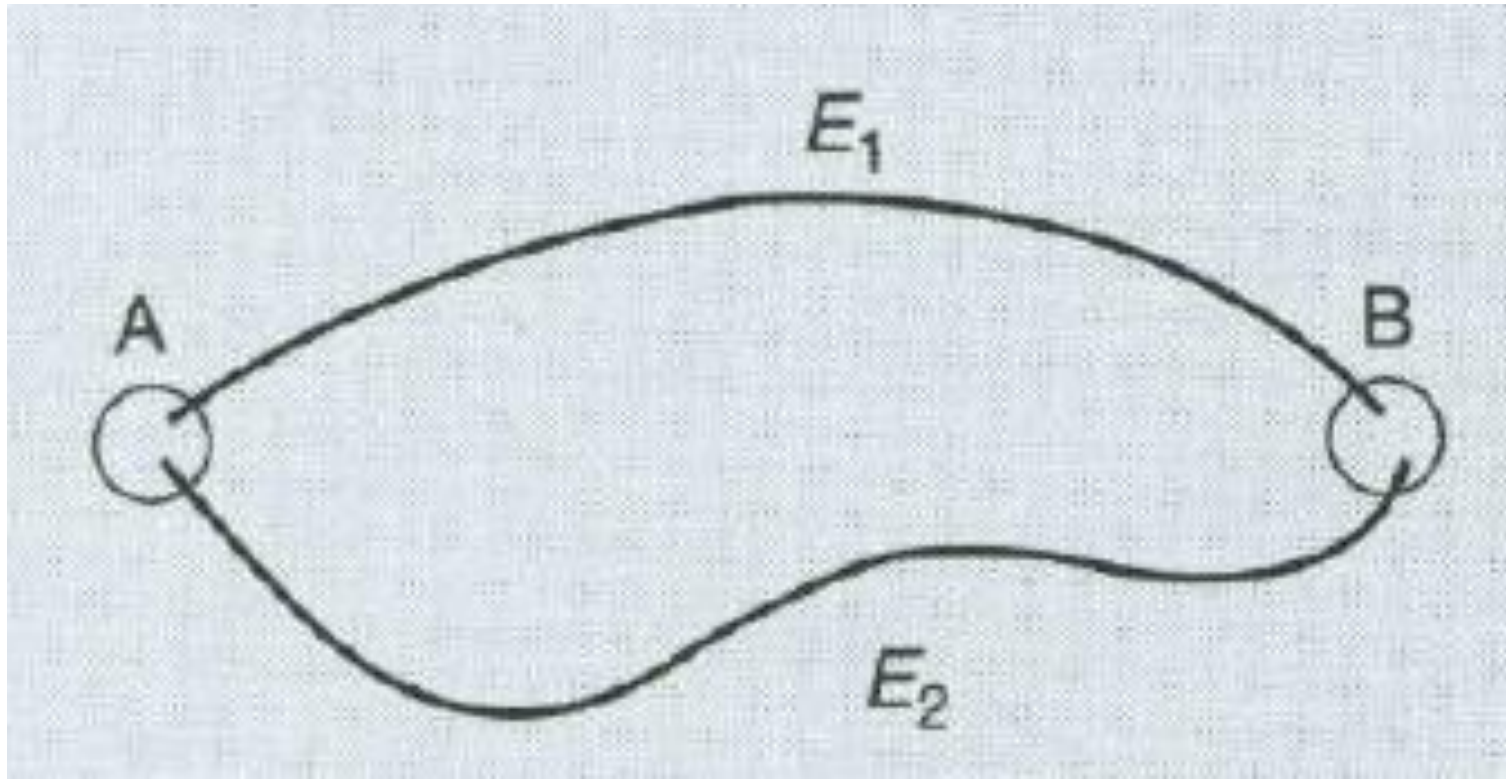
$$= E_1' \cap E_2'$$

Task: 1

Suppose there are two highways from city A to city B as shown in fig. Let E_1 represent the event that Route 1 is open and E_2 that Route 2 is open to traffic.

At least one of the two routes is open means:

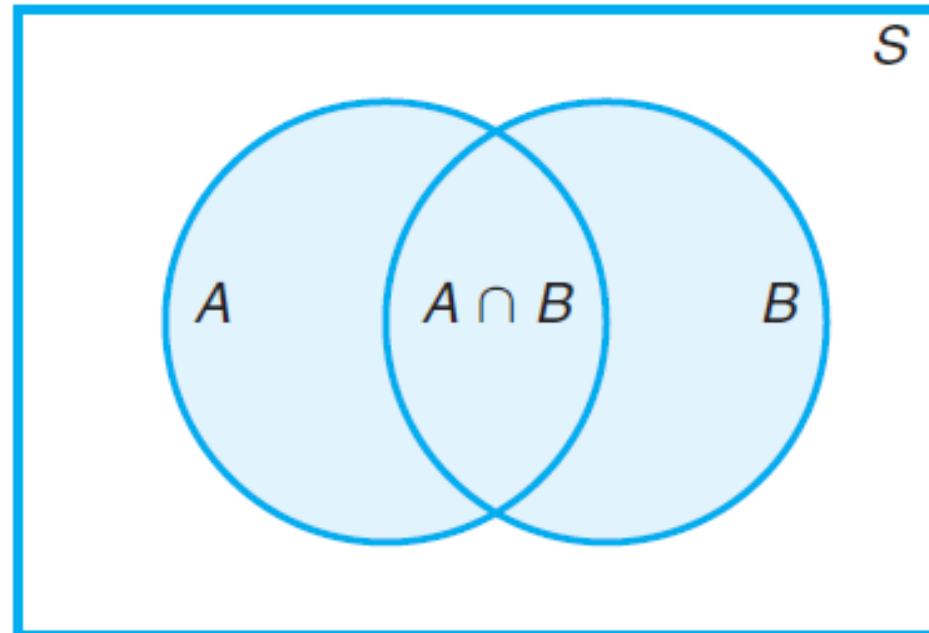
Either Route 1 open or Route 2 open = $E_1 \cup E_2$



Additive Rule

If A and B are two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$



Corollary

If A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B).$$

If $A_1, A_2, \dots, A_n$ are mutually exclusive, then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n).$$

If $A_1, A_2, \dots, A_n$ is a partition of sample space S , then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n) = P(S) = 1.$$

For three events A , B , and C ,

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) \\ &\quad - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C). \end{aligned}$$

Example: 4

Q: What is the probability of getting a total of 7 or 11?

$$P(A) = \frac{n(A)}{n(S)}$$

$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6),$
 $(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6),$
 $(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6),$
 $(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6),$
 $(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6),$
 $(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$

$$n(S) = 36$$

Total of 7 = $\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$

$$n(7) = 6$$

Total of 11 = $\{(5, 6), (6, 5)\}$

$$n(11) = 2$$

$$P(A \cup B) = P(A) + P(B)$$

$$P(7 \text{ or } 11) = P(7) + P(11)$$

$$= \frac{6}{36} + \frac{2}{36}$$

Example: 5

Q.1: A statistics class for engineers consist of 25 industrial, 10 mechanical, 10 electrical and 8 civil engineering students. If a person is randomly selected by the instructor to answer a question, find the probability that the student chosen is

(a) an industrial engineering major

Sol: Total engineering students: $n(ES) = 53$

No of industrial engineering students: $n(IE) = 25$

$P(\text{Student is an industrial engineering major}) = P(IE) =$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(IE) = \frac{n(IE)}{n(ES)} = \frac{25}{53} = 0.47$$

Example: 5

Q.1: A statistics class for engineers consist of 25 industrial, 10 mechanical, 10 electrical and 8 civil engineering students. If a person is randomly selected by the instructor to answer a question, **find the probability that the student chosen is a civil engineering or an electrical engineering major.**

Sol: Total engineering students: $n(ES) = 53$

No of civil engineering students: $n(CE) = 8$

No of electrical engineering students: $n(E) = 10$

$P(\text{Student is a civil or electrical engineering major}) =$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \cup B) = P(A) + P(B)$$

$$P(CE \cup EE) = P(CE) + P(EE)$$

$$= \frac{n(CE)}{n(ES)} + \frac{n(EE)}{n(ES)}$$

$$= \frac{8}{53} + \frac{10}{53} = 0.34$$

Example: 6

John is going to graduate from an industrial engineering department in a university by the end of the semester. After being interviewed at two companies he likes, he assesses that his probability of getting an offer from company A is 0.8, and his probability of getting an offer from company B is 0.6. If he believes that the probability that he will get offers from both companies is 0.5, what is the probability that he will get at least one offer from these two companies?

$P(A) = 0.8$
$P(B) = 0.6$
$P(A \cap B) = 0.5$

$P(\text{At least one offer}) = P(A \cup B)$
$= P(A) + P(B) - P(A \cap B)$
$= 0.8 + 0.6 - 0.5$
$= 0.9$

Task

A contractor is starting two new projects-jobs 1 and 2. There is some uncertainty on the scheduled completion of each job; at the end of 1 year, the condition of completing each job may be defined as follows:

- a. C= definitely completed
- b. Q= completion questionable
- c. I= definitely incomplete

- 1) Describe the sample space regarding the completion of both jobs.
- 2) What is the probability that **at least one job will definitely be completed** at the end of 1 year (A)? (Assume that state of completion of both jobs is equally likely)
- 3) Find the prob. that only one of the two projects will be completed (B)?

Task

Q.1: It is common in many industrial areas to use a filling machine to fill boxes full of product. This occurs in the food industry as well as other areas in which the product is used in the home, for example, detergent. These machines are not perfect, and indeed they may A , fill to specification, B , under-fill, and C , overfill. Generally, the practice of under-filling is that which one hopes to avoid. Let $P(B) = 0.001$ while $P(A) = 0.990$.

- (a) Give $P(C)$.
- (b) What is the probability that the machine does not under-fill?
- (c) What is the probability that the machine either overfills or under-fills?

Task

Q.2: According to *Consumer Digest* (July/August 1996), the probable location of personal computers (PC) in the home is as follows:

Adult bedroom: 0.03

Child bedroom: 0.15

Other bedroom: 0.14

Office or den: 0.40

Other rooms: 0.28

(a) What is the probability that a PC is in a bedroom?

(b) What is the probability that it is not in a bedroom?

Task

The major airline industry is subject to labor strikes by the Pilots (C), Mechanics (M), and the Flight attendants (F) or by two or more of these labor groups. Assuming the strikes by the three individual labor groups to be statistically independent, find the probability of a labor strike in the major airline industry in next 3 years. Respective probabilities of strike by the three individual groups are: $P(C)=0.03$, $P(M)=0.05$, $P(F)=0.05$.

علامہ اقبالؒ

نہیں ڈرتے دلاورِ رنج و غم سے
نہ پرواہ ہے اُنھیں کچھ آسماں کی
بہائیں بھی اگر آنسو کبھی وہ
علامت جائے سوزِ نہاں کی